

Ecolight Maestro

Time & Light Intensity Based Street Light Automation using LPC2129 ARM7

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Abstract

Conventional street lighting systems commonly remain switched on for fixed durations irrespective of actual ambient conditions, resulting in avoidable energy wastage. Ecolight Maestro is an embedded automation system, built around the LPC2129 ARM7 microcontroller, that addresses this inefficiency by combining real-time clock (RTC) based scheduling with ambient light intensity sensing through a Light Dependent Resistor (LDR) and the on-chip ADC. Street lights are switched ON only when both conditions confirm night-time operation — the current time is at or beyond a configured evening threshold (18:00) and the measured light intensity falls below a set ADC value — and are switched OFF otherwise, ensuring lights are never left on unnecessarily during daylight.

The system provides a 16x2 LCD for continuous status monitoring — displaying time, date, day, and ADC readings — and a 4x4 matrix keypad, accessed through an EINT0 external interrupt, that lets the user open a menu to update RTC parameters (hour, minute, second, day, date, month, and year) on demand. The design was validated through circuit simulation in Proteus and developed using Embedded C in Keil uVision, demonstrating correct dual-condition decision logic, responsive interrupt-driven menu access, and stable RTC and ADC peripheral integration.

By coupling time-based scheduling with real-time environmental sensing, Ecolight Maestro offers a low-cost, energy-efficient, and user-configurable approach to automated street lighting, with clear extensibility toward IoT-based remote monitoring, GSM fault reporting, solar power integration, and adaptive brightness control in future iterations.